

Yuan Gao

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Education

- University of Cambridge** — PhD in Computer Science 2023 – Present
- Structured probabilistic modeling for graph-conditioned language generation, focusing on probabilistic learning/inference (e.g., EM, variational Bayes, gradient-based optimization) and extending methods from semantic graphs to general graphs.
- University of Cambridge** — MPhil in Advanced Computer Science, *with Distinction* 2021-2022
- Thesis: User Simulator for Engaging Conversations on Wikipedia. Developed knowledge-grounded dialogue generation with RL-style training and benchmarked automatic dialogue metrics.
- Carnegie Mellon University** — B.S. in Computer Science, *University Honors*, Minor in Math 2017-2021

Skills

- Structured ML:** structured generative modeling, latent-variable models, EM, variational Bayes
- NLP:** transformers, knowledge-grounded dialogue generation, RLHF, reward modeling, evaluation
- Systems:** compilers, runtime estimation, CI/test automation
- Programming:** C++, Python

Selected Publications

- P1: A Computational Simulation of Language Production in First Language Acquisition.**
Yuan Gao, Weiwei Sun. EMNLP 2025. [paper] [code] [talk]
- P2: BLiSS: Evaluating Bilingual Learner Competence in Second Language Small Language Models.**
Yuan Gao et al.. BabyLM Workshop @ EMNLP 2025. [paper] [code] [huggingface]
- P3: EtymoLink: A Structured English Etymology Dataset.**
Yuan Gao, Weiwei Sun. LChange @ ACL 2024. [paper] [code]

Experiences

- University of Cambridge** — Research Assistant Nov 2022 – Jan 2023
- Built an RLHF-style pipeline for content generation for Cambridge Assessment, combining prompting and evaluation for knowledge-grounded outputs. Enabled rapid iteration for an adaptive learning prototype.
- Salesforce** — Software Engineer Intern Jun 2021 – Aug 2021
- Implemented new SAQL/SQL capabilities for Tableau / Salesforce Analytics Cloud across parser → compiler → optimizer → distributed execution. Shipped customer-facing math functions and distributed query support to unblock analytics use cases.
- Carnegie Mellon University** — Research Assistant, DialRC Lab Sep 2020 – Mar 2021
- Trained seq2seq/transformer user simulators and fine-tuned DialoGPT-style dialogue models; generated interaction data for RL-based training and evaluation. Enabled scalable offline experiments for dialogue policy learning and metric-driven evaluation.
- Salesforce** — Software Engineer Intern Jun 2020 – Aug 2020
- Developed an ML-based runtime estimator for Einstein Discovery and integrated it end-to-end across backend + frontend (Python/Java/JavaScript). Improved training-time predictability and production readiness via model comparison, tuning, and deployment-focused validation.
- Carnegie Mellon University** — Research Assistant Jun 2019 – Apr 2020
- Simulated cryo-electron tomograms and evaluated DoG particle-picking parameters; contributed modules to AITom (open-source tomography toolkit). Improved reproducibility of experimental pipelines and accelerated parameter exploration for downstream analysis.
- Mastercard** — Software Engineer Intern Jun 2018 – Aug 2018
- Added multi-language support for Pay with Rewards automated testing using Java (Maven, Spring DI) and optimized the end-to-end workflow. Reduced regression runtime from 4 hours to <45 minutes.